Cover Letter

I am a **Software Architect**, a **Database Architect**, and a seasoned **Software Engineer** with *expertise* in C and Go languages, databases, Cloud Services, Distributed Systems, Postgres RDBMS internals, Kubernetes, and Docker containers.

Since 2010, I have kept pace with frontend technologies as well, so things like NodeJS, TypeScript, various Frontend frameworks, etc. are also in my wheelhouse.

Some ***notable achievements*** of my career are: Developed Amazon Aurora Postgres's transaction logging system, developed Postgres-as-a-Service 3 times, once at EnterpriseDB, then at GE Digital on AWS and on Kubernetes.

Please take the time to read at least the first few pages of my resume, and I assure you, ***you won't be disappointed*** :)

The resume is organized as follows:

1. Cover Letter
   This page. You're reading the cover letter.

2. One-page resume
   Page 2. High-level overview of my skills and strengths.

3. Open Source contributions
   Page 3. Details of my contributions to various Open Source projects. I'm proud of each of the items on this page.

4. Summary/list of the all the jobs I've held over my career.
   Page 4.

5. Detailed work experience
   Page 6. The details of all the work I've accomplished over my career.

Gurjeet Singh

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full resume: http://gurjeet.singh.im/resume

- Software Architect, Database Architect, Systems Software Engineer with 20+ years of experience.
- Proficient in C, PostgreSQL database internals, GNU/Linux, and Windows.
- Proficient in Docker Containers, Kubernetes, AWS, Azure, GCP.
- Fair exposure to C++, NodeJS, GoLang, Java, Oracle, MS-SQL Server, Solaris OS.
- Engineering leader with a strong sense of ownership in everything I do.

I have spent most of my career developing RDBMS internals, including my work on Oracle's Cost Based Query Optimizer, Amazon Aurora Postgres, and notable contributions to the Postgres open source project; for details, see the "Detailed Work Experience" section in my full resume, linked above.

I have spoken at and presented at various Conferences; see the "Speaking Engagements" section.

I have over 8 years of experience as a Postgres Database Architect, DBA, Database Trainer, and Database Consultant. In these roles, among other things, I helped customers learn Postgres development, Postgres internals, performance tuning, migrated applications from other databases to Postgres, and recovering data from Postgres corruptions.

Qualifications:

1. Agile/Scrum Master
When: January, 2015
Institution: Scrum Alliance

2. Post Graduate Diploma in Advanced Computing
Duration: Feb 2001-July 2001
Institution: C-DAC (Center for Development of Advanced Computing)

3. Bachelor of Engineering (Mechanical)
Duration: 1996-2000
Institution: Gulbarga University, India.

Fields of Interest:

OS internals, Compilers, RDBMS, Distributed Systems, Linux Containers, Kubernetes, RISC-V.

Skills:

Programming Languages: C, Golang, C++, Java, Javascript, Typescript, Shell, and some assembly.

Operating Systems: Windows 9x, Windows NT, Sun Solaris, GNU/Linux.

Databases: Oracle, MS-SQL server, Sybase, SQLite, and an expert in Postgres.

Web Technologies: NodeJS, AngularJS, REST APIs, Microservices, XML, HTML, Web 2.0, PHP.

Open Source Contributions

Duration: 2005 – now

Projects:

Postgres RDBMS

Language: *C*

Role: *Contributor, Hacker, Developer, Reviewer*

Description: Postgres is the most advanced Open Source RDBMS

Contributions to Core Postgres:

- Found *and* fixed a Security Vulnerability (CVE-2009-4136) (commit, discussion, docs)
- Feature: Replace Primary Key index of a table (commit, submission, discussion, docs)
- Feature: Improve interaction with Linux OOM Killer (commit, submission, discussion, docs)
  It was very difficult to convince Tom Lane that this idea had merit.
- Feature: The \ir command for psql (commit, submission, review, docs)
- Fix a Windows OS-specific race condition (commit, submission)
- Feature: Improve pgbench to support large databases (commit, submission, discussion)
- Feature: pgbench: add client_id variable (commit, submission)
- List of all my contributions to core (many not mentioned in the list above): link to search results

Contributions to Postgres ecosystem

- ***Postgres Tools***
  - Creator of Index Advisor extension (Git repo)
  - Creator of Postgres Hibernator extension (Git repo)
  - Creator of DBYardstick – a Lightweight TPC-C Benchmark in NodeJS (Git repo)
  - Creator of pg_plan_guarantee Extension – (Git repo)
- ***Postgres-Hackers*** mailing-list contributions/reviews/conversations
  - link to search results
- ***Blogs***
  - Current Blog, Random Neural Firings, Left-Brain Dump

Speaking Engagements at Conferences

- ***Postgres Open***
  2015 - Postgres Inside Out
  2014 - How Databases Work - For Developers, Accidental DBAs and Managers
  2013 - How Postgres is different from (read BETTER than) your database
  2011 - Introducing Postgres Enterprise Manager
  2011 - Why you should NOT move away from Oracle
  2011 - Postgres Plus Cloud Database
- ***PGCon***(the Original, Postgres Hackers' annual conference)
  2011 - Best Practices with Ora2Pg (link)

Other Open Source Projects, and Work

- Creator of the vim-darwin package in NixOS (merged Pull Request)
- Creator of ZFS File System driver for Docker containers (Git repo)

Work Experience (summary)

0. ***Open Source***
   Designation: Contributor / Hacker / Developer
   Duration: 2005 – Now

   Description: Contributed new features and bug-fixes to Postgres project. Developed projects and Postgres extensions of my own. Speaker at conferences. See "Open Source Contributions" section for details.

1. ***Amazon Web Services***
   Designation: Senior Software Engineer
   Date: Jun 2022

   Description: Contribute new features and bug fixes to the Postgres open source project.

2. ***Supabase Inc.***
   Designation: Fellow Engineer
   Date: Sep. 2021

   Description: Contribute new features and bug fixes to Postgres community, and initiate & work on company's long-term, multi-year projects.

3. ***GE Digital***
   Designation: Principal Engineer
   Date: May 2016

   Description: Architected, developed, owned, and operated Postgres-as-a-Service at Predix Platform on AWS, as well as air-gapped Kubernetes platform. Service Owner of Blobstore service.

4. ***Amazon Web Services***
   Designation: Software Development Engineer
   Date: May 2015

   Description: Architect and develop Amazon Aurora for Postgres. Help with Amazon RDS Postgres.

5. ***Thales e-Security, San Jose, CA., USA***
   Designation: Senior Engineer, Database Specialist
   Date: Nov. 2014

   Description: Design and develop an HSM (Hardware Security Module) Manager using service-oriented architecture. Also, migrate a long-standing market-leading hardware key-manager from embedded MySQL to Postgres.

6. ***EnterpriseDB Corporation, Boston, MA., USA***
   Designation: Database Architect
   Date: Feb. 2011

   Designation: Principal Systems Engineer

Date: Feb. 2010

Description: Worked on database internals and contributed to Open Source Postgres database project. My work as a primary/sole contributor includes Index Advisor, Reindex Primary Key Indexes, Postgres Hibernator, DBYardstick Benchmark development. I also co-lead the Postgres Plus Cloud Database and Postgres Enterprise Manager projects.

I provided consulting services, where I helped customers and partners perform Architectural Health Checks of Postgres databases, perform database migrations, conduct trainings, develop custom replication solutions to other database, etc. while also participate in Postgres community contributing code patches and features.

7. ***EnterpriseDB Software India Pvt. Ltd., Pune, India.***
   Designation: Lead Engineer.
   Date: July 2006

   Description: Researching new features and implementing them in the EnterpriseDB RDBMS (which is based on Postgres). Act as Trainer, Consultant and Remote-DBA for customers and partners, on different occasions.

8. ***CommVault Systems (India) Pvt. Ltd., Hyderabad, India.***
   Designation: Software Engineer.
   Date: May 2005

   Description: Develop and bug-fix the Report Generation Server, developed in C++, which interacts with MS-SQL Server for its data storage needs.

9. ***Oracle India Private Ltd., Bangalore, India.***
   Designation: Member Technical Staff.
   Date: Nov 2003

   Description: Suggesting and implementing new features, and fixing defects in the *Cost Based Optimizer* module of the Oracle RDBMS, on Solaris and Linux platforms.

10. ***Aztec Software and Services Ltd., Bangalore, India.***
    Designation: Developer.
    Date: Sept 2003

    Description: Designed *External Stored Procedures* feature for an RDBMS product, which was being developed for a customer.

11. ***Vaman Technologies (R&D) Ltd, Mumbai, India.***
    Designation: Junior Software Engineer.
    Date: July 2001

    Description: Worked on various components of an RDBMS, including the TPC-C benchmarking utility, the Query Parser, Procedural Language support, OS independent kernel, and the OS abstraction layer.